YEAR 7 · ICE ENGINEERS 冰雪工程师 · WEEK 1 OF 10 · 40 MINUTES

What do scientists do? 科学家做什么?

Scientists separate what they see (observations) from what they think it means (inferences), and change their story when new evidence appears.

- I can tell the difference between an observation and an inference.
- I can change my inference when I get new evidence.

scientist 科学家 · observation 观察 · inference 推断 · evidence 证据 · clue 线索 · footprint 脚印 · track 足迹

THE LESSON

Skills check	15	Seat groups by plan; hand out skills check and read stems aloud. Silent, alone, no help.
Survey and Today slide	3	Collect checks; read survey aloud, students tick faces. Today slide (1 min): goals, thumbs.
Hook	3	Stage 1 snow tracks: 'Who walked here?' Pair guesses in any language; introduce the series.
Learn	4	Melted snowman: observation or inference? Thumbs on 4 statements; model the two frames.
Do	10	Footprint puzzle in groups: Stage 2 at about 3 min, Stage 3 at about 6. Ask 'How do you know?'
Think and Reflect	5	Two Reporters share; 'Can we be 100% sure?'; exit ticket; first logbook line; collect both.

Hands-on: Footprint puzzle (paper): groups open 3 reveal cards stage by stage, recording observations and inferences.

Collect or assess: Baseline skills check (out of 20) and interest survey into the tracker for Week 10; exit ticket; first logbook page.

SAFETY

- No practical hazards beyond normal classroom rules.
- Keep bags and coats under the tables so walkways stay clear.

WATCH OUT

- Skills check: full 15 minutes, no help; never go through or return papers (same paper in Week 10).
- An inference is not a guess: link it to an observation with 'because'. Short of time? Skip Stage 2.

Language: Frames: 'I observe ___.' 'I infer ___ because ___.' Chinese talk welcome; report in English. 推断 is not 猜 (guess).

Next week: Week 2 Do Now: mystery cups. This week make 12 cups, print and cut question sort cards, find a hardback book.

PREP FOR ONE CLASS

before the lesson about 187 min (154 could be done by someone else); after about 108 min (some times estimated)

2 OR MORE WEEKS BEFORE

- [ ] Order Week 1 kit: A4 card, envelopes, pouches, A3, name cards, pencils, box 15 min · technician
- [ ] Book reprographics: 24 logbooks, 28 dictionaries (A5, 90 gsm) 10 min · teacher

THE WEEK BEFORE

- [ ] Print, laminate and cut role cards: 6 sets of 6 (6 A4 pouches) 25 min · technician
- [ ] Print 30 language cards double-sided and laminate (30 A5 pouches) 30 min · technician
- [ ] Print 18 puzzle cards on card, skills checks, surveys; make 24 name cards 30 min · technician
- [ ] Print 24 worksheets double-sided; check exit ticket cut line first 5 min · reprographics
- [ ] Write seating plan, set up tracker tab, print mark scheme and kit card 23 min · teacher
- [ ] Optional, if not using card: print puzzle cards on paper, laminate (18 A4) optional 20 min · technician
- [ ] Put up 3 A3 posters and A4 frame poster; print, cut 6 sets of frame cards 35 min · technician

THE DAY BEFORE

- [ ] Label 18 stage envelopes, 1 puzzle card in each; Stages 2 and 3 for desk 5 min · technician

ON THE DAY, BEFORE THE LESSON

- [ ] Set out name cards, role cards, pencils; papers and envelopes on desk 8 min · technician

AFTER THE LESSON

- [ ] Check envelope sets; enter scores in tracker; file papers, do not return 28 min · technician
- [ ] Read exit tickets; mark skills checks out of 20 within the week 80 min · teacher
- [ ] Optional: enter question-by-question marks on Question_analysis tab optional 15 min · technician

PRINT

- Skills check x24 (3 sheets each, double-sided, last side blank)
- Interest survey x24 (half-A4, 2 per A4 sheet, cut in half: 12 sheets)
- Worksheet x24 (2 sheets each, double-sided)
- Teacher copies: kit card, skills check mark scheme (2 pages)
- Puzzle cards x6 sets of 3 (18 A4 card, teacher note on back of Card 3)
- Role cards x6 sets (1 A4 sheet each, laminate and cut)
- Classroom language card x30 (A5, double-sided, laminate)
- Posters: 3 A3 language posters, 1 A4 sentence frame poster
- Sentence frame cards x6 sets of 16 (24 A4 sheets, cut)
- Reprographics booklets: logbook x24, dictionary x28 (A5)

GET OR MAKE (NEW THIS LESSON)

- A4 white card x18 sheets (puzzle cards)
- C4 envelopes x18 (puzzle stages)
- A4 laminating pouches x6 (role cards)
- A5 laminating pouches x30 (language cards)
- A4 pouches x18 for paper puzzle cards (optional)
- A3 paper x3 sheets (posters)
- Name cards x24
- HB pencils x24, plus spares on the teacher desk
- Class box with lid x1
- Class assessment folder x1 (if not already there)
- Laminator and guillotine or scissors (access)
- Rulers for the tables (optional)
- Puzzle envelope sets x6 (make)
- Group seating plan with starting roles (make)

Have ready from earlier weeks: Nothing.

YEAR 7 · ICE ENGINEERS 冰雪工程师 · WEEK 2 OF 10 · 40 MINUTES

Thought experiments 思想实验

Scientists can test an idea in their heads first (a thought experiment), then turn their curiosity into questions an experiment can answer.

- I can tell the difference between a testable and a not testable question.
- I can write a testable question and a prediction with a reason.

thought experiment 思想实验 · **prediction** 预测 · **reason** 理由 · **testable** 可检验的 · **evidence** 证据 · **fall** 下落 · **gravity** 重力 · **air resistance** 空气阻力

THE LESSON

Do Now	4	Shake, listen and weigh 2 sealed mystery cups; write observation and inference; Today slide.
Hook	3	Vote: heavy or light ball, which lands first? Predict with 'because'; start Galileo story.
Learn	7	Galileo storyboard; teacher book and paper drops, predict each; testable questions.
Do	18	Card sort: testable or not (8 min); worksheet: 3 testable questions with predictions (10 min).
Think and Reflect	8	Heavier things fall faster: always, sometimes, never? Pair talk, share, exit ticket, logbook.

Hands-on: Groups handle sealed mystery cups and sort question cards; only the teacher does the book and paper drops.

Collect or assess: Card sort checked in class (note groups with 3+ errors); 3 testable winter questions (Rubric strand 1); exit ticket.

SAFETY

- Choking risk: mystery cups stay taped shut and are never opened; nothing in mouths.
- Falling book: teacher only, from about 1 m onto table or floor; hands and feet clear.

WATCH OUT

- Heavier is not always faster: drop flat and crumpled sheets (same mass) and insist on 'sometimes'.
- Never reveal cup contents; keep the kit card (A to L key) on the teacher desk, not on tables.

Language: Band 1 sits by the language buddy. Traps: 'light' (not heavy), 'fall' (verb, not autumn), 'testable' (not an exam).

Next week: Week 3 spinners: wing length vs fall time. This week: print spinner card, mark 2.0 m and 1.8 m on walls.

PREP FOR ONE CLASS

before the lesson about 67 min (57 could be done by someone else); after about 26 min (some times estimated)

2 OR MORE WEEKS BEFORE

- [ ] Order or gather cup contents, 24 opaque paper cups, masking tape, marker 10 min · technician
- [ ] Book printing: worksheets x24, A3 Galileo poster, kit card (A5) 5 min · reprographics
- [ ] Optional: add 6 A4 Galileo storyboard table copies to the print job optional reprographics

THE WEEK BEFORE

- [ ] Make 12 mystery cups: object in, second cup on top, tape, label A to L 15 min · technician
- [ ] Print sort cards on A4 white card: 6 sets x 2 sheets = 12 sheets 5 min · reprographics
- [ ] Cut out sort cards: 6 sets x 14 cards = 84 cards (guillotine) 15 min · technician
- [ ] Put each sort card set (12 questions + 2 headers) in an envelope 5 min · technician
- [ ] Find a hardback book; practise the paper-on-top drop once 5 min · teacher

THE DAY BEFORE

- [ ] Put up the A3 Galileo storyboard poster 2 min · technician

ON THE DAY, BEFORE THE LESSON

- [ ] Set out 2 cups per table, role and frame cards, pencils; teacher desk kit 5 min · teacher

AFTER THE LESSON

- [ ] Clean-up: sort cards counted into envelopes, cups and logbooks to boxes 3 min · students
- [ ] Check cup tape; re-tape any loose cup before the next class 3 min · technician
- [ ] Read exit tickets: who needs the Do Now recap, who is ready for Stretch 10 min · teacher
- [ ] Record a language band (1, 2 or 3) per student in tracker column G 10 min · teacher

PRINT

- Worksheet x24 (4 pages, 2 sheets each, double-sided, 48 sheets)
- Galileo storyboard poster x1 (A3 landscape, single-sided)
- Question sort cards x6 sets (2 sheets A4 card each, 12 sheets)
- Galileo storyboard table copies x6 (A4, single-sided) (optional)
- Teacher copy: kit card with mystery cup key (A5)

GET OR MAKE (NEW THIS LESSON)

- Paper cups, opaque x24
- Cup contents: marbles x3, cotton ball, coin, paper clip
- Cup contents: eraser, dice, small bell, bottle cap
- Cup contents: small wooden block, dry rice (1 teaspoon)
- Masking tape x1 roll, marker pen x1
- Small envelopes x6 (for sort card sets)
- Hardback book, about A4 size x1
- Plain A4 paper x3 sheets (drop demo)
- Teacher storage box for the mystery cups x1
- Mystery cups x12, sealed, labelled A to L (make)

Have ready from earlier weeks: Class box, Scientist's Logbooks x24, Pencils x24, Group role card sets x6, Sentence frame card sets x6 (check), Classroom language cards x24, My Science Dictionary x24

YEAR 7 · ICE ENGINEERS 冰雪工程师 · WEEK 3 OF 10 · 40 MINUTES

Meet the variables 认识变量

In a fair test we change only one thing, measure one thing, and keep everything else the same.

- I can name the independent, dependent and control variables in an experiment.
- I can explain why a fair test changes only one thing.

variable 变量 **independent variable 自变量** **dependent variable 因变量** **control variable 控制变量** **fair test 公平测试** **air resistance 空气阻力** **spinner 纸直升机** **wing length 翼长**

THE LESSON

Do Now	4	Two paper planes picture: 'Is this a fair race?' answered with the frame; Today slide.
Hook	3	Teacher drops 4 cm and 8 cm spinners together from 2 m; class votes which lands first.
Learn	7	'I change / I measure / I keep the same' with gestures and vocab panel; cloze; safety slide.
Do	18	Make three spinners (4, 6, 8 cm), drop from 2 m, time, record; picture sort, variables table.
Think and Reflect	8	Odd one out: which test is not fair? Pair talk, share with a frame; exit ticket.

Hands-on: Groups cut and fold three card spinners (4, 6, 8 cm wings), drop each from a 2 m wall mark and time the fall.

Collect or assess: Variables table (Core D) and results (Core C); exit ticket: IV, DV and one control variable, collected at the door.

SAFETY

- Falls: no standing on chairs or tables; Dropper stands flat (1.8 m mark if needed).
- Scissors points down, cut away from fingers; never unbend or point paper clips.
- Tripping: keep bags off the floor under the drop marks.

WATCH OUT

- Check drop height as groups drift lower; 1.8 m mark only if no one reaches 2 m, and then for every drop.
- IV and DV mixed up, or DV called the thing kept the same: go back to the hand gestures.

Language: Frames: 'It is / is not fair because...', 'I change, I measure, I keep the same'. Chinese welcome; Band 1 beside buddy.

Next week: Keep spinners in labelled envelopes and leave wall marks up: Week 4 adds repeats, means and anomalies.

PREP FOR ONE CLASS

before the lesson about 57 min (45 could be done by someone else); after about 20 min (some times estimated)

THE WEEK BEFORE

- [ ] Send worksheet to reprographics: 24 copies, 72 sheets, double-sided, B&W 5 min · reprographics
- [ ] Print 9 spinner sheets on 160 gsm card at 100%; check the 1 cm square 5 min · technician
- [ ] Print 6 odd-one-out sheets (single-sided), cut each into A6 cards A to D 10 min · technician
- [ ] Print one kit card (A5) for the teacher 2 min · teacher
- [ ] Tape 2.0 m and 1.8 m wall marks at 6 stations, away from windows and vents 10 min · technician
- [ ] Make demo S (4 cm) and L (8 cm) spinners, one clip each; test from 2 m 10 min · teacher

THE DAY BEFORE

- [ ] Label six A5 envelopes with the group names 5 min · technician

ON THE DAY, BEFORE THE LESSON

- [ ] Set out 6 kit trays and teacher desk; check wall marks and clear floor 10 min · technician

AFTER THE LESSON

- [ ] Check envelopes hold S, M, L spinners; store with logbooks in class box 5 min · teacher
- [ ] Count back scissors and stopwatches, collect card sets, return kit, tidy 5 min · technician
- [ ] Read exit tickets to plan Week 4 Do Now recap and spot who is Stretch-ready 10 min · teacher

PRINT

- Worksheet x24 (3 sheets each, double-sided, B&W)
- Spinner template, 160 gsm card x9 sheets (at 100%, includes 3 spares)
- Odd-one-out sheet x6 (1 A4 sheet each, cut into 4 A6 cards)
- Teacher copy: kit card (A5)

GET OR MAKE (NEW THIS LESSON)

- Paper clips x26 (28 mm, all the same size)
- Scissors x12 pairs
- Stopwatches x6
- Metre sticks x6
- Kit trays x6 (requisition lists 1: decide)
- A5 envelopes x6
- Masking tape x1 roll (Week 2 roll may do)
- Make: demo spinners x2 (S 4 cm, L 8 cm wings)
- Make: wall drop marks at 6 stations (2.0 m and 1.8 m)

Have ready from earlier weeks: Marker pen (Week 2), Pencils x24 (Week 1), Group role cards x6 sets, Classroom language cards x30, Sentence frame card sets x6, Language wall posters x4, Logbooks x24 in the class box, My Science Dictionary x28 (optional)

YEAR 7 · ICE ENGINEERS 冰雪工程师 · WEEK 4 OF 10 · 40 MINUTES

Measure it well 学会准确测量

Measurements always vary a little, so scientists repeat them, work out a mean and look carefully at any result that does not fit.

- I can record repeat measurements with the correct units and calculate a mean.
- I can spot an anomaly and explain why I leave it out of the mean.

measure 测量 · **unit** 单位 · **second (s)** 秒 · **repeat** 重复 · **mean** 平均值 · **anomaly** 异常值 · **results table** 结果表格 · **reaction time** 反应时间

THE LESSON

Do Now	4	Retrieval: IV, DV and a control; why two people timed one drop differently. Today slide.
Hook	3	Teacher drops one L spinner from 2 m; three volunteers time it. Why are the times different?
Learn	7	Units, repeats, mean = total ÷ number of values, anomaly; worked example; students try Core C.
Do	18	Groups time 3 repeats per wing length (9 drops), circle anomalies with reasons, find means.
Think and Reflect	8	'Include it or leave it out?' pair talk, share with a frame; exit ticket; logbook line.

Hands-on: Groups drop their Week 3 S, M and L spinners from the 2 m (or 1.8 m) mark, 3 repeats each, stopwatch timed.

Collect or assess: Group results table with means and circled anomalies (kept for Week 5); exit ticket: circle anomaly, mean without it.

SAFETY

- Falls: never stand on chairs or tables; Dropper on the floor (1.8 m mark if needed).
- Tripping: keep bags off the floor under the drop marks.
- Scissors: carry points down. Paper clips: never unbend or point at anyone.

WATCH OUT

- Check every group divides by 2, not 3, after leaving out an anomaly; circle anomalies, never rub out.
- Short of time: same Measurer, 9 drops in about 8 min; else two repeats or the sample data card.

Language: Band 1 beside a Band 3 buddy; Chinese welcome in pair talk. Traps: mean = 平均值 (not unkind), second = 秒, Repeat 1 = 第1次.

Next week: Week 5 graphs these means and has a 10-minute checkpoint: keep the envelopes and the 6 sample data cards.

PREP FOR ONE CLASS

before the lesson about 40 min (28 could be done by someone else); after about 18 min (some times estimated)

2 OR MORE WEEKS BEFORE

- [ ] Book printing: worksheets x24 (double-sided, collated), kit card 5 min · reprographics

THE WEEK BEFORE

- [ ] Print 3 spare spinner sheets, 160 gsm card at 100% (unless done in Week 3) 3 min · reprographics
- [ ] Print 6 sample data cards (A5 landscape, single-sided) 2 min · technician
- [ ] Laminate the 6 sample data cards in A5 pouches (optional) optional 5 min · technician
- [ ] Check envelope spinners and wall marks (2.0 m); rebuild or re-press 13 min · technician

THE DAY BEFORE

- [ ] Collect and check 7 stopwatches and tray, scissors, clips, metre stick 5 min · technician
- [ ] Test-drop the demo L spinner from 2 m; rebuild if bent 2 min · teacher
- [ ] Plan roles (new Measurer, Dropper) and Band 1 / Band 3 buddy pairs 5 min · teacher

ON THE DAY, BEFORE THE LESSON

- [ ] Set out kit on tables and teacher desk; clear bags from under drop marks 5 min · teacher

AFTER THE LESSON

- [ ] Clean-up: spinners and results table in envelope, kit counted back 3 min · students
- [ ] Store envelopes, check results tables, note unusable data; keep cards 5 min · teacher
- [ ] Read exit tickets; note who needs a recap before the Week 5 checkpoint 10 min · teacher
- [ ] Record freezer decision; Week 6: canteen ice (Plan A) or thermometers (B) teacher

PRINT

- Worksheet x24 (5 pages, 3 sheets each, double-sided)
- Sample data cards x6 (A5 landscape, single-sided)
- Spare spinner templates x3 (A4 160 gsm card, 100% scale)
- Teacher copy: kit card (A5)

GET OR MAKE (NEW THIS LESSON)

- Stopwatches x7 (1 per group, 1 extra for Hook) and tray
- Scissors x6 pairs (repairs only)
- Paper clips, 28 mm x12 (spares)
- Metre stick x1
- A5 laminating pouches x6 (optional)

Have ready from earlier weeks: Spinner envelopes x6 (18 spinners), Demo L spinner (Week 3), Wall drop marks, 6 stations, Masking tape roll (Week 3), Role cards and frame cards, 6 sets each, Logbooks x24 in the class box, Science dictionaries x24, Pencils x24

YEAR 7 · ICE ENGINEERS 冰雪工程师 · WEEK 5 OF 10 · 40 MINUTES

Spot the pattern + Checkpoint 发现规律 + 阶段检测

A graph turns a table of numbers into a picture, so we can spot the pattern and write a conclusion that uses the data as evidence.

- I can choose between a bar chart and a line graph and plot my results accurately.
- I can write a conclusion using 'As the ___ increases, the ___ ___.'

graph 图表 | bar chart 条形图 | line graph 折线图 | axis (axes) 坐标轴 | plot 描点 | line of best fit 最佳拟合线 | pattern 规律 | conclusion 结论 | increase 增加

THE LESSON

Do Now	3	Mean of 1.40, 1.46, 1.52 s (1.46 s) and units; hand out the group envelopes.
Checkpoint	10	Alone, in silence; Chinese answers accepted; no help; collect at exactly 10 minutes.
Hook	2	Same means as a bar chart and a line graph: which tells the story better? Vote with a reason.
Learn	5	Words or groups: bar; numbers: line. IV on x-axis, DV on y-axis; crosses; line of best fit.
Do	14	Copy means (own or sample card), label axes, plot, line of best fit, conclusion frame.
Think and Reflect	6	Vote BAR or LINE on cards A to D with the frame; exit ticket and logbook line; collect sheets.

Hands-on: No practical: after the checkpoint, students plot the Week 4 spinner means (or sample data) and conclude.

Collect or assess: Checkpoint out of 12 (into tracker column Q); graph and conclusion (Rubric strand 4); exit ticket: bar or line? (bar)

SAFETY

- No practical hazards beyond normal classroom rules.
- Carry rulers and pencils safely; do not wave them near faces.

WATCH OUT

- Checkpoint overruns: collect at exactly 10 minutes and count papers against the register.
- Sample data card Mean column is blank: groups work out means, leaving out the 1.12 s anomaly at 8 cm.

Language: Frame: 'As the ___ increases, the ___ ___.' Band 1 sits with the language buddy; Chinese accepted everywhere.

Next week: Week 6 starts the ice project: ice cube rescue (Plan A) or keep it warm (Plan B), using the freezer decision.

PREP FOR ONE CLASS

Plan A: before the lesson about 40 min (23 could be done by someone else); after about 119 min (some times estimated)

Plan B: before the lesson about 40 min (23 could be done by someone else); after about 129 min (some times estimated)

THE WEEK BEFORE

- [ ] Print worksheets x24 (grid at 100%), checkpoints x25, cards A to D x1 set 13 min · reprographics
- [ ] Print and cut 24 BAR | LINE voting cards (test sheet first) or reuse a set 5 min · technician
- [ ] Print teacher copies: kit card, mark scheme, lesson plan, handover forms 5 min · teacher

THE DAY BEFORE

- [ ] Check group envelopes; note which groups need the sample data card 5 min · teacher
- [ ] Sharpen 24 pencils plus spares; spares and sharpener on teacher desk 5 min · technician

ON THE DAY, BEFORE THE LESSON

- [ ] Set out kit; incoming teacher's seat at the back with plan and checkpoint 7 min · teacher

AFTER THE LESSON

- [ ] Clean-up: checkpoints counted first; logbooks to box; keep 2 spinner sets 4 min · students
- [ ] Mark checkpoints out of 12 (no half marks); enter in tracker column Q 40 min · teacher
- [ ] Read graphs, conclusions and exit tickets; note who struggled with axes 15 min · teacher
- [ ] Complete one handover note per class; book the 30-minute Week 6 walkthrough 22 min · teacher
- [ ] Strip trial: make 12 x 2 cm block, fold 5 foil troughs; build, clear 1 rig 23 min · technician
- [ ] Strip trial (Plan B): allergy check, mix salt dough, weigh 5 balls of 18 g (Plan B) 10 min · technician
- [ ] Strip trial (Plan A): freeze 5 troughs of 12 ml for 24 h+, test 3, keep 2 (Plan A) 15 min · teacher
- [ ] Strip trial (Plan B): press balls into troughs, dry a week, test 3, keep 2 (Plan B) 15 min · teacher

PRINT

- Worksheet x24 (2 sheets each, double-sided, grid at 100%)
- Checkpoint paper x25 (1 for incoming teacher; 1 sheet, double-sided)
- BAR | LINE voting cards x24 (3 A4 card sheets, double-sided, cut)
- Bar or line? cards A to D, 1 set (4 A5 sheets, single-sided)
- Teacher copies: kit card, mark scheme, lesson plan, handover note form

GET OR MAKE (NEW THIS LESSON)

- 30 cm rulers x24
- Pencil sharpener x1
- Kitchen foil x5 pieces, about 20 x 10 cm (strip trial)
- Template block 12 x 2 cm (wood offcut or taped rulers)
- Foil troughs x5 and plain trial strips x5 (make)
- Salt dough balls x5, 18 g (flour, salt) (Plan B)
- Kitchen scale, 1 g (Plan B)
- Measuring cylinder, 25 ml (or 20 ml syringe) x1
- Freezer space for 5 troughs, 24 h+ (Plan A)
- Dry shelf for 5 strips, one week (Plan B)
- Textbooks x6 (two stacks of three)
- Thin plastic bags x2 (Plan A)
- Carrier or freezer bag with handles x1, string
- Plastic cup x1 and marker pen (25 ml pour cup)
- Bucket or deep washing-up bowl x1
- Jug, about 2 litres x1

Have ready from earlier weeks: Class box, Logbooks x24, Group envelopes x6 (spinners, results), Role card sets x6, Sentence frame card sets x6, Science Dictionaries x24, Sample data cards x6 (Week 4), Class pencils x24, sharpened

YEAR 7 · ICE ENGINEERS 冰雪工程师 · WEEK 6 OF 10 · 40 MINUTES

Ice cube rescue 冰块保卫战

A thermal insulator slows heat transfer: a coat on a snowman slows heat moving from the warm air into the snow, so it melts more slowly.

- I can plan and carry out a fair test to compare thermal insulators.
- I can explain, using my results, why a coat slows down melting (or cooling).

thermal insulator 隔热材料 · insulate 隔热 · heat transfer 热传递 · melt 熔化 · meltwater 融化的水 · temperature 温度 · thermometer 温度计 · material 材料 · concept cartoon 概念漫画

THE LESSON

Do Now	4	A: wrap 3 bagged cubes, bag A bare, note start time. B: wrap cups B to D, A bare. Goals.
Hook	3	Snowman cartoon vote: 'I agree with ___ because ___.' B: teacher pours 150 ml warm water.
Learn	7	Heat moves warm to cold; a coat slows it. Fair test, cup A control; safety. B: start readings.
Do part 1	14	Hands off cups. Variables table, prediction, Stretch, Challenge. B: end readings at 15 min.
Think 1 (B: Do continues)	4	A: groups choose a question, draft planning frame page 1. B: finish temperature drops, ranks.
Measure (B: frame page 1)	4	A: gloves on, unwrap, measure each bag's meltwater; never cut. B: planning frame page 1.
Think and Reflect	4	Reporters: best and worst material; 'Who was right?' (Owen); exit ticket; frames into folders.

Hands-on: A: bagged ice in 3 materials, 1 bare control; meltwater after about 29 min. B: 50 °C water, drop over 15 min.

Collect or assess: Worksheet write-up (Rubric strands 2 to 4); planning frame page 1; exit ticket: best insulator, number and unit, reason.

SAFETY

- Wet floor: cups on trays, mop spills at once. No eating or drinking; wash hands.
- Plan A: cold skin: gloves for ice held over a minute; ice stays in sealed bags.
- Plan B: only teacher pours water (about 50 °C, max 55 °C); never mercury thermometers

WATCH OUT

- Start the ice (A) or cup jackets (B) in the Do Now, trays out first. A: never cut the measuring slot.
- Misconceptions: 'coats make heat', 'cold comes in'. Heat moves, cold does not; shiny does not mean best.

Language: Frame: 'I agree with ___ because ___.' Chinese talk welcome; Chinese answers count. 'Insulator' here means heat.

Next week: Frames stay in folders. Now: strip trial, buy untreated sawdust. A: book freezer; B: flour, salt, drying shelf

PREP FOR ONE CLASS

Plan A: before the lesson about 80 min (60 could be done by someone else); after about 28 min (some times estimated)

Plan B: before the lesson about 75 min (55 could be done by someone else); after about 30 min (some times estimated)

2 OR MORE WEEKS BEFORE

- [ ] Book freezer or canteen ice (about 28 cubes); check 2 ice trays, cool box — Plan A · 5 min · teacher
- [ ] Reserve 7 thermometers (never mercury), 6 stopwatches, kettle, 2 jugs — Plan B · 10 min · technician

THE WEEK BEFORE

- [ ] Gather kit, 6 folders; laminate and put up A3 cartoon; label 24 cups A to D — 25 min · technician
- [ ] Send print job for your plan only (designs rebuilt after the language pass) — 5 min · reprographics
- [ ] Gather 28 zip-lock bags, 12 pairs of gloves, 6 measuring cylinders — Plan A · 5 min · technician
- [ ] Mark 150 ml line on 24 cups; check 7 thermometers agree within 1 °C — Plan B · 10 min · technician

THE DAY BEFORE

- [ ] Set project groups (tracker col F); label 6 folders, add frames; buddies — 10 min · teacher
- [ ] Freeze 2 trays of ice (about 28 cubes); cut 15 x 15 cm squares, pack trays — Plan A · 15 min · technician
- [ ] Cut 25 x 10 cm jackets and 10 x 10 cm slit lids; pack 6 trays — Plan B · 5 min · technician

ON THE DAY, BEFORE THE LESSON

- [ ] Bag, seal 28 ice cubes; trays on tables, folders on desk; cool box at door — Plan A · 15 min · technician
- [ ] Trays out, folders on desk; boil kettle; Do Now: jug to about 50 °C, 3.6 l — Plan B · 10 min · teacher

AFTER THE LESSON

- [ ] Clean-up by roles (Plan B: teacher first counts the 7 thermometers, 2 min) — 3 min · students
- [ ] Dry and store reusable materials; mop if needed; restock the 6 trays — 10 min · technician
- [ ] Keep frames in folders for Week 7; read exit tickets; note leaders, support — 15 min · teacher

PRINT

- Worksheet, Plan A or B, x24 (3 sheets each, double-sided)
- Week 7 planning frame, Plan A or B, x8 (3 sheets each, double-sided)
- Concept cartoon x1 (A3, single-sided, then laminated)
- Teacher copy: kit card for your plan (A5, single-sided)

GET OR MAKE (NEW THIS LESSON)

- Trays x6, group folders x6, A3 laminating pouch x1
- Paper towels x36 sheets; mop and bucket (or a roll)
- Scissors and a ruler (teacher prep)
- Foil, cotton wool sheet, bubble wrap, newspaper, fabric
- Cups labelled A to D x24 (Plan B: 150 ml line marked)
- Plan A: 18 pre-cut 15 x 15 cm squares (6 sets of 3)
- Plan A: plastic cups x30, elastic bands x30
- Plan A: ice cubes x28, each sealed in a zip-lock bag
- Plan A: ice-cube trays x2, cool box x1, freezer drawer
- Plan A: disposable gloves x12 pairs
- Plan A: measuring cylinders x6 (25 ml or 10 ml)
- Plan B: 36 pre-cut pieces (18 jackets, 18 slit lids)
- Plan B: identical cups (about 250 ml) x25, bands x36
- Plan B: thermometers x7 (spirit or digital, no mercury)
- Plan B: kettle, 2 jugs (3.6 litres at about 50 °C)
- Plan B: measuring cylinder x1 (to mark the 150 ml line)

Have ready from earlier weeks: Class box with 24 logbooks, Role cards x6 sets (with buddy card), Sentence frame card sets x6, Classroom language cards x24, Science Dictionaries x24, Pencils x24; marker pen x1, Plan B: stopwatches x6, Plan A: stopwatches x6 (optional)

Chinese: draft, needs native-speaker check

YEAR 7 · ICE ENGINEERS 冰雪工程师 · WEEK 7 OF 10 · 40 MINUTES

The ice ship mission 冰船计划

Fibres make a brittle material into a composite that is harder to break; a good plan lets us test this fairly.

- I can plan a fair test: question, variables, prediction with a reason, method and risks.
- I can make identical samples that change only one thing, and label them clearly.

composite material 复合材料 · reinforce 加固 · fibre 纤维 · brittle 脆的 · strength 强度 · sawdust 木屑 · mould 模具 · sample 样品 · hazard 危险源 · risk 风险

THE LESSON

Do Now	4	Folders back; Week 6 retrieval; Lead Scientist reads plan page 1; today's goals. Slides 1 to 2.
Hook	3	Pykrete story: Pyke's wartime ice ship; predict why sawdust makes ice stronger. Slides 3 to 4.
Learn	6	How fibres stop cracks; questions, tests; frame tour; model a risk row; Core A. Slides 5 to 10.
Do part 1 (plan)	13	Groups finish frame in roles; teacher approves samples on the board; Core B, Stretch. Slide 11.
Think and Reflect	4	Plan swap with checklist and one fix; teacher signs 'Plan approved'; exit ticket. Slide 12.
Do part 2 (make samples)	10	Safety slide; pairs make and label about 9 strips per group; 2 min tidy. Slides 13 to 17.

Hands-on: Pairs make, label about 9 fibre strips per group in foil troughs: ice (Plan A) or salt dough (Plan B).

Collect or assess: Approved planning frame (photographed); samples logged on class board; exit ticket: sample code, a control with reason.

SAFETY

- Both: untreated sawdust only, never MDF. Plan A: keep damp, water first, never tip
- Plan A: mop spills at once; only the teacher carries trays and uses the freezer
- Plan B: allergic students never touch dough; never eat it; knead in bag; wash hands

WATCH OUT

- Every group includes the control (plain ice or 0 g) and changes one thing; check the board first.
- Strip trial done before this lesson; labels in pencil, never cut labelling; trays flat and level.

Language: Frame: 'I predict ___ because the fibres ___.' Band 1: buddy, Core B first. Mould is the container, not fungus.

Next week: Week 8 test day: samples from freezer or shelf. Get books, string, bags, jugs, trays, towels, gloves, mop.

PREP FOR ONE CLASS

Plan A: before the lesson about 120 min (95 could be done by someone else); after about 27 min (some times estimated)

Plan B: before the lesson about 130 min (110 could be done by someone else); after about 35 min (some times estimated)

2 OR MORE WEEKS BEFORE

- [ ] Check handover note for Plan A or B; send that plan's print job
  5 min · reprographics

THE WEEK BEFORE

- [ ] Buy fibres, foil, bags, paper cups, tape, kitchen roll; Plan B: flour, salt
  15 min · technician
- [ ] Plan A: book a freezer drawer to Week 8; clear, label, level it day before Plan A 10 min · teacher
- [ ] Plan B: check allergy list; give allergic students a non-dough role
  Plan B 5 min · teacher
- [ ] Read Week 6 drafts, list portions; day before, put up board, trial decision 15 min · teacher
- [ ] Fold 60 foil troughs on Week 5 block; cut 64 labels, 10/group; prep fibres 45 min · technician
- [ ] Laminate the A3 class project board (optional) optional
  5 min · technician

THE DAY BEFORE

- [ ] Plan A: pre-portion and label fibres (up to 36); sawdust damp, sealed
  Plan A 20 min · technician
- [ ] Plan B: pre-portion and label fibres (up to 36); clear a dry shelf
  Plan B 15 min · technician

ON THE DAY, BEFORE THE LESSON

- [ ] Plan B: make 2 dough batches; weigh 54 balls of 18 g plus spares, bag each Plan B 20 min · technician
- [ ] Set out 6 kit trays and group folders (Plan A: mop ready)
  10 min · technician

AFTER THE LESSON

- [ ] Plan A: carry trays flat to the freezer; freeze at least 24 hours
  Plan A 5 min · teacher
- [ ] Plan B: trays flat to drying shelf; on day 3 peel foil, turn strips
  Plan B 13 min · teacher
- [ ] Store kit and spare fibre; photograph approved plans; read exit tickets 20 min · teacher

PRINT

- Worksheet, Plan A or B x24 (2 A4 sheets each; page 3 single-sided)
- Sample labels x2 A4 sheets (64 labels, single-sided)
- Class project board, Plan A or B x1 (A3, single-sided)
- Teacher copy: kit card for your plan x1 (A5, double-sided)

GET OR MAKE (NEW THIS LESSON)

- 25 ml cylinders or 20 ml syringes x12 (Plan A)
- Small mixing cups x18, water cups (200 ml) x6 (Plan A)
- Teaspoons x18
- Flat trays for 9 troughs x6 (Week 6 trays if flat)
- Untreated sawdust, never MDF (up to 30 g, per plans)
- Tissue, shredded paper, cotton wool (up to 12 g each)
- Kitchen foil x60 pieces, about 20 cm x 10 cm
- Small paper cups for fibre portions x36
- Small zip-lock bags x54 for dough balls (Plan B)
- Labelled bags for spare fibre (1 per fibre type)
- Paper towels x24 sheets
- Masking tape (6 rolls on requisition)
- Plain flour 500 g and salt 580 g (Plan B)
- Mixing bowl, cup measure, damp cloth (Plan B)
- Freezer drawer (Plan A) or dry shelf (Plan B)
- A3 laminating pouch x1 (optional)

Have ready from earlier weeks: Group folders x6, planning frames, 2 spare planning frames, Role card sets x6, Sentence frame card sets x6, Science Dictionaries x24, Class box; logbooks x24 (optional), Pencils x24, marker pen, Scale, template block; Plan A jug, mop

Test day 测试日

A test is only as good as how it is done: test every sample the same way, record every result, and repeat to see how much results vary.

- I can carry out a strength (or melting) test safely, in exactly the same way for every sample.
- I can record repeat results with units and notice anything unusual.

strength 强度 · **load 负载** · **snap 断裂** · **volume 体积** · **millilitre (ml) 毫升** · **mass 质量** · **repeat 重复** · **anomaly 异常值** · **catch tray 接水盘**

THE LESSON

Do Now	3	Samples, folders out; Safety Officer reads hazards, Recorder copies codes, others build 2 rigs.
Hook	2	Plain or fibre sample: which holds more water, and how will each break? Hands-up prediction.
Learn	6	Demo rig (10 cm span, 25 ml pours, 5 s wait), recording rule, roles, safety. Melt test starts.
Do	22	2 rigs per group, samples in mixed order; record ml and how it broke; 3 repeats (min 2).
Think and Reflect	7	Results onto class board; discuss 'Anomaly or not?'; exit ticket; tidy, mop, wash hands.

Hands-on: Groups break-test Week 7 strips on 2 water-bag rigs. Plan A: up to 2 groups melt-test (15 min). Plan B: dough.

Collect or assess: Results table with repeats and notes (Rubric strand 3); group results on class board (photo for Week 9); exit ticket.

SAFETY

- Bag falls on snap: bucket under each bag; hands, feet clear; pourer kneels to side.
- Wet floor (biggest risk): towels on tables, mop by door; mop spills and pause group.
- Plan A: gloves for ice. Plan B: never eat dough; flour allergy: pour or record only.

WATCH OUT

- Same test each time: 10 cm span, centre loop, gentle 25 ml pours, no pressing; Plan A: test within 1 min.
- Record every result, even early snaps, with how it broke; push for 3 repeats (accept 2).

Language: Chinese talk welcome; write in English (Chinese notes count). Traps: load is not loud, snap = break, volume = liquid.

Next week: Week 9 pools class data: photograph the board, keep results sheets in folders, note groups missing data.

PREP FOR ONE CLASS

Plan A: before the lesson about 129 min (105 could be done by someone else); after about 27 min (some times estimated)

Plan B: before the lesson about 111 min (85 could be done by someone else); after about 27 min (some times estimated)

2 OR MORE WEEKS BEFORE

- [ ] Keep 2 spare trial strips for the Hook (slides show 1 plain, 1 fibre) 2 min · teacher

THE WEEK BEFORE

- [ ] Book printing: worksheet x24 (3 sheets each, double-sided, stapled) Plan A 10 min · reprographics
- [ ] Book printing: worksheet x24 (2 sheets each); check exit ticket cut first Plan B 10 min · reprographics
- [ ] Collect rig kit for 6 groups and demo (books, bags, buckets, jugs, towels) 30 min · technician
- [ ] Collect cool boxes, gloves, bags, melting kit; laminate 2 station cards Plan A 20 min · technician
- [ ] If cool boxes are short, bubble-wrap 6 lidded boxes optional Plan A 15 min · technician
- [ ] Tie 13 strings, mark 12 cups at 25 ml; print kit card and rig diagrams 32 min · technician

THE DAY BEFORE

- [ ] Check handover note and board (plan, melting groups, trial); fill folders 10 min · teacher
- [ ] Check dough strips feel hard; check the flour allergy list Plan B 7 min · teacher

ON THE DAY, BEFORE THE LESSON

- [ ] Set out 6 kit trays (2 rigs each, books beside), mop and bucket by door 15 min · technician
- [ ] Set up the demo rig at the front (both plans) 5 min · teacher
- [ ] Just before: move ice strip trays from freezer into insulated boxes Plan A 5 min · teacher

AFTER THE LESSON

- [ ] Photograph class board; check results sheets in folders, read exit tickets 12 min · teacher
- [ ] Clean up: dry books, rinse buckets and jugs, dry towels, mop, return kit 15 min · technician

PRINT

- Worksheet, Plan A x24 (3 sheets each, double-sided, stapled)
- Worksheet, Plan B x24 (2 sheets each, double-sided, stapled)
- Rig diagram A4 x6 (1 sheet each, laminated)
- Rig diagram A3 x1 (board copy)
- Melting station card A5 x2 (laminated, Plan A)
- Kit card for your plan, A5 double-sided (1, or 1 per group)

GET OR MAKE (NEW THIS LESSON)

- Textbooks x78 (students' own are fine)
- Carrier or freezer bags with handles x13
- Buckets or deep bowls x13 (catch trays)
- String x13 lengths (pre-tied hanger strings)
- Plastic cups x12, 25 ml cylinder, marker (pour cups)
- Jugs (about 2 litres) x6, waste bowls x6
- Rulers (30 cm) x7
- Towels x13, paper towels (1 roll), mop and bucket
- Tap water, about 12 litres
- Laminating pouches A4 x6, A5 x2 (A5 for Plan A)
- Plan A: cool boxes with lids x6
- Plan A: thin plastic bags x12, gloves x18 pairs
- Plan A: 10 ml cylinder x2, lids x20, stopwatch x2
- Plan A: lidded boxes x6 + bubble wrap (optional)

Have ready from earlier weeks: Ice strip trays x6 (Plan A, freezer), Dough strip trays x6 (Plan B, dry shelf), 2 spare trial strips (Hook), Group folders + planning frames x6, Role card sets x6, Sentence frame card sets x6, Class project board, Kit trays x6

Chinese: draft, needs native-speaker check

YEAR 7 · ICE ENGINEERS 冰雪工程师 · WEEK 9 OF 10 · 40 MINUTES

What does it mean? 数据说明了什么?

Pooled class results are stronger evidence than one group; the right graph shows the pattern, and CER plus an honest evaluation say how sure we are.

- I can choose the right graph and write a conclusion: claim, number evidence, science reasoning.
- I can explain one problem with our test, how it affected the results, and how sure we are.

pool (data) 汇总（数据） **class mean 全班平均值** **bar chart 条形图** **line graph 折线图** **claim 主张** **evidence 证据** **reasoning 推理** **evaluate 评价** **improvement 改进** **confident 有把握的**

THE LESSON

Do Now	4	Folders back; finish Week 8 group means (no anomalies), check against class table; Today slide
Hook	3	'Snow Owls 292 ml, Ice Foxes 300 ml: who is right?' Pair talk: trust one group or the class?
Learn	7	Pooling (class mean = mean of group means); bar or line vote; teacher models CER; Core A and B
Do	18	Panels by role: Core C, Stretch, Challenge, panel card, glue on poster; first glued by min 12
Think and Reflect	8	Self-assessment slip (4 min); split the Week 10 talk with talk frame card; exit ticket; collect

Hands-on: No practical: groups pool class data, choose a graph, and write, cut out and glue panels onto an A3 poster.

Collect or assess: Poster: evidence for strands 1 to 5; worksheets show who wrote what; self-assessment slips; exit ticket: talk parts.

SAFETY

- No practical hazards beyond normal classroom rules.
- Scissors for the panel cards: carry scissors with the points down.

WATCH OUT

- Own group's result is not 'the answer': class mean is better evidence; a word variable needs a bar chart.
- Time: have the results table ready; if short, do Panels 1, 4, 5 today and 6, 7 at the start of Week 10.

Language: Chinese talk welcome; panels in English or Chinese with English key words. Traps: pool 汇总, claim 主张, mean 平均值.

Next week: Keep posters flat, talk cards in folders and exit tickets; posters go up in talk order for the Week 10 Expo

PREP FOR ONE CLASS

before the lesson about 75 min (45 could be done by someone else); after about 51 min (some times estimated)

2 OR MORE WEEKS BEFORE

- [ ] Book Week 9 printing with reprographics (part of the Week 7 booking) 5 min · teacher

THE WEEK BEFORE

- [ ] Print A3 posters, panel cards (100%), pooling sheets, slips, talk cards 10 min · reprographics
- [ ] Print worksheets (48 sheets) and kit card, from the language-pass designs 5 min · reprographics
- [ ] Cut self-assessment slips in half (24) and fold the 24 talk frame cards 10 min · technician
- [ ] Print and laminate sample data cards (3 per plan or only your plan's 3) 10 min · technician
- [ ] Write results table on A3 from Week 8 photo (Plan A: add melting rows) 15 min · teacher
- [ ] Choose a flat, safe place to store the A3 posters overnight 5 min · teacher

THE DAY BEFORE

- [ ] Put missing Week 8 results sheets in folders; pick groups for sample data 5 min · teacher

ON THE DAY, BEFORE THE LESSON

- [ ] Set out group kits, role and frame cards, sample cards, A3 table, slips 10 min · technician

AFTER THE LESSON

- [ ] Clean-up: posters flat in folders, slips collected, glue capped and counted 3 min · students
- [ ] Store posters flat; file worksheets and self-assessment slips 3 min · teacher
- [ ] Read every poster; pencil in draft levels for strands 1 to 5 in the tracker 30 min · teacher
- [ ] Compare slips with your judgement; note gaps of 2+ levels for a chat 10 min · teacher
- [ ] Read exit tickets: every group member has a talk part in poster order 5 min · teacher

PRINT

- Worksheet x24 (2 sheets each, double-sided)
- Pooling sheet x6 (1 per group, A4 landscape, double-sided)
- Pooling sheet side 1 enlarged to A3 x1 (board table)
- Poster template x7 (A3, single-sided; 6 groups plus 1 spare)
- Panel card set x7 (4 sheets each, single-sided, print at 100%)
- Self-assessment slip x24 (A5, 2 per sheet: 12 sheets, cut in half)
- Talk frame card x24 (1 sheet each, double-sided, folded)
- Sample data cards, laminated: Plan A x3 (double-sided), Plan B x3
- Teacher copies: kit card (A5, 2 sides), sample data answers (optional)

GET OR MAKE (NEW THIS LESSON)

- Laminating pouches A4 x6 (3 if only your plan's cards)
- Glue sticks x12 (2 per group)
- Thick black felt-tip pens x12 (2 per group)
- Thin black felt-tip pens x12 (2 per group)
- Class results table on the A3 board copy (make)

Have ready from earlier weeks: Group folders x6 (Week 7 and 8 sheets), Role card sets x6 (with language buddy), Sentence frame card sets x6, My Science Dictionary x24, Pencils x24, Scissors x6 (1 per group), 30 cm rulers x12 (2 per group), Logbooks x24 in class box (optional)

YEAR 7 · ICE ENGINEERS 冰雪工程师 · WEEK 10 OF 10 · 40 MINUTES

Ice Engineers Expo 冰雪工程师成果展

Scientists share what they found, show their evidence, answer questions honestly and listen to feedback; that is how science gets checked.

- I can present my part of our investigation clearly in English, using key words and our numbers.
- I can ask or answer a question about evidence, and give kind, specific feedback.

Expo 成果展 · **presentation** 展示 · **audience** 观众 · **evidence** 证据 · **conclusion** 结论 · **feedback** 反馈 · **confident** 有把握的 · **scientist** 科学家

THE LESSON

Do Now (skills check)	15	Final skills check, same paper and conditions as Week 1; all papers collected at minute 15.
Hook	1	The Expo is open: order of talks and who asks whom; get talk frame, notes, card, slip ready.
Do (Expo talks)	15	2-minute talk per group at its poster; next group asks a card question; Expo notes, slips.
Think and Reflect	4	Pair talk: 'How do scientists find things out?'; feedback slips to groups; exit ticket.
Survey and certificates	5	Survey read aloud (3 min); thank the class; certificates or when and where; gallery walk.

Hands-on: No practical: 2-minute Expo talks at A3 posters (ice strips Plan A, dough strips Plan B); one question each.

Collect or assess: Skills check (column S, growth); survey (U to Z); strand 6 from talk record, strands 1 to 5 from posters (AC to AH).

SAFETY

- No practical hazards beyond normal classroom rules.
- Posters on walls or boards: keep the walkways in front of them clear of bags.

WATCH OUT

- Never shorten the skills check or skip the survey; if short of time, use 90-second talks and 3 questions.
- Fix talk order (next group asks; first asks last) and label each feedback slip with the group asked.

Language: Answer: 'Our evidence shows ___, but it does not show ___.' Band 1: buddy reads Chinese frame. Chinese counts for Core.

Next week: Last lesson: gallery walk, then store posters; mark and enter results; certificates, reports within a week.

PREP FOR ONE CLASS

Plan A: before the lesson about 66 min (21 could be done by someone else); after about 310 min (some times estimated)

Plan B: before the lesson about 66 min (21 could be done by someone else); after about 305 min (some times estimated)

2 OR MORE WEEKS BEFORE

- [ ] Book printing with reprographics, incl. cards and certificates on card 5 min · teacher

THE WEEK BEFORE

- [ ] Print skills checks, surveys and rest of print job; cut surveys and slips 20 min · reprographics
- [ ] Plan room layout, sticky notes; book gallery walk and certificate ceremony 10 min · teacher

THE DAY BEFORE

- [ ] Fix talk order; fill Slides 3, 5, 10, talk record; print cards; label slips 15 min · teacher
- [ ] Only if given today: print certificates fixed by Week 9 levels (optional) optional 15 min · teacher

ON THE DAY, BEFORE THE LESSON

- [ ] Put posters up in talk order; set out papers, table kit, clipboard, timer 15 min · teacher

AFTER THE LESSON

- [ ] Collect and sort work; mark skills checks; enter scores and surveys 60 min · teacher
- [ ] Finalise strands 1 to 6; apply award rules; print and present certificates 60 min · teacher
- [ ] Read Expo notes and exit tickets; write and print 24 skills reports 90 min · teacher
- [ ] Return Week 1 and 10 skills checks after marking, if wanted (optional) optional 5 min · teacher
- [ ] Gallery walk signs up; then photograph, store posters; store and return kit 45 min · technician
- [ ] Programme summary; return or file logbooks; three lines for next year 45 min · teacher
- [ ] Plan A close-out: clear freezer shelves of leftover samples Plan A 10 min · technician
- [ ] Plan B close-out: throw away leftover dough and strips Plan B 5 min · technician

PRINT

- Skills check x24 (1 sheet each, double-sided, Week 1 master)
- Interest survey x24 (half-A4, 12 sheets, cut in half)
- Expo notes worksheet x24 (2 sheets each, double-sided)
- Peer feedback slips x24 (A6, 6 sheets, cut in four)
- Question cards x24 (2 sheets, A4 card if possible, cut)
- Gallery walk signs x2 (1 sheet each)
- Certificates x25 (A4 card: 24 awards + 1 Most Improved)
- Skills reports x24 (1 sheet each, after the lesson)
- Teacher copies: run sheet, talk record (2 sheets), kit card (A5)
- Skills check mark scheme, if Week 1 copy not kept (optional)

GET OR MAKE (NEW THIS LESSON)

- Pencils x2 and string x2 (gallery walk signs)
- Sticky notes x2 pads
- Clipboard x1
- Poster wall fixings (masking tape, putty or pins)
- Storage box labelled 'Ice Engineers' x1
- Guillotine or scissors x1
- Camera or phone x1 (school device)

Have ready from earlier weeks: A3 group posters x6 (Week 9), Group folders x6 with talk frames x24, Class box with logbooks x24, Role cards x6 sets (language buddy), Science dictionaries x24, Pencils x24, glue sticks (2 per group), Week 9 self-assessment slips x24, Week 1 skills checks x24 (optional)

Chinese: draft, needs native-speaker check